BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: 112101

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 20

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. $2 \sin \pi/4 \cos \pi/4 =$ _____.

(A) 2
(B) 1/2
(C) √2
(D) 1

Q2. $n^2 > n + 3$ for all n _____.

(A) >2
(B) = 1
(C) >2
(D) >1

Q3. The period of tan (x) = _____.

(A) 2π
(B) π/2
(C) π
(D) -π

Q4. |2 x; 3 6| = 0, then x = _____.

(A) -3
(B) 4
(C) 3
(D) -4

Q5. cos (2π ± θ) = _____.

(A) -cos θ
(B) cos θ
(C) sin θ
(D) ±cos θ

Q6. $\text{Sin}^{-1}(-1/2)$ = _____.

(A) -2π
(B) [illegible in source scan]
(C) π
(D) 3π

Q7. Let A = {-1,0,1} and B = {-1,0,1}, which one is a 1-1 function from A to B _____.

(A) x^3 (B) $|x|$

(C) x^2 (D) $\sqrt{x}$

Q8. (n, 0) = _____.

(A) n! (B) 1

(C) n (D) n-1

Q9. If vector a = xi + yj ≠ 0 vector then |vector a| = _____.

(A) > 0 (B) ≥ 0

(C) < 0 (D) None of these

Q10. If tan β < 0, then β lies in _____ quadrant.

(A) I or II (B) III or IV

(C) II or III (D) II or IV

Q11. Magnitude of Z = (0,-2) is.

(A) -2 (B) (0,2)

(C) 2 (D) None of these

Q12. i(1/i - i) = _____.

(A) -1 (B) -i

(C) 1+i (D) 2

Q13. A and B are two sure events, then P(A ∪ B) = _____.

(A) 0 (B) 1

(C) 1/2 (D) 2

Q14. i·j = _____.

(A) k (B) 1

(C) -k (D) 0

Q15. A harmonic sequence is the reciprocal of the terms of _____ sequence,

(A) Constant (B) Geometric

(C) Pascal (D) Arithmetic

Q16. If vector a = 2i - 3j + 4k and vector b = i + 3j - 2k then a·b = _____.

(A) -15 (B) [illegible in source scan]

(C) 15 (D) 0

Q17. If $a > 0$, then the parabola $f(x) = ax^2 + bx + c$ opens _____.

(A) Left (B) Right

(C) Downward (D) Upward

Q18. The inverse of the linear function $f(x) = 2x + 5$ is _____ function.

(A) Rational (B) Linear

(C) Constant (D) Quadratic

Q19. ${}^8P_2$ = _____.

(A) 28 (B) 56

(C) 16 (D) None of these

Q20. The solution region of $y - x > 4$ contains the point.

(A) (0,5) (B) (0,3)

(C) (5,0) (D) None of these

SECTION "B"

Time Allowed: 2 hours and 40 minutes | **Maximum Marks:** 50

QI. Attempt any Ten Parts out of the following. Each Part carries equal marks.

(i) Express $(4+i)/(3+5i)$ in the form $a + b\,i$.

(ii) Let vector $a = i + 3j - 4k$ and vector $b = 2i - 3j + 5k$. Find the value of m so that vector a + m vector b is orthogonal to vector b.

(iii) Convert to reduced Echelon form the 2×2 matrix shown in the paper.

(iv) The second term of an HP is 1/2 and 5th term is 1/4. Find the 12th term.

(v) Find the value of 'k' if 2k+7, 6k-2, 8k-4 are in A.P. Also find the sequence.

(vi) Prove by mathematical induction that $1.2+2.3+3.4+ \ldots + n(n+1) = n(n+1)(n+2)/3$.

(vii) A bag contains 4 white, 5 red and 6 green balls. 3 balls are drawn at random. What is the probability that all balls are white.

(viii) Find the inverse function of $f(x) = x/3 - 5$.

(ix) Show that $\sin(\alpha+\beta)/(\cos\alpha \cos\beta) = \tan(\alpha) + \tan(\beta)$.

(x) Solve the triangle with dimensions $\alpha=35°$, $\beta=70°$, $c=115$.

(xi) Find the length of the vector AB from the point A(-3,5) to B(7,9). Also find a unit vector in the direction of AB.

(xii) Draw the graph of $y = -4 + \sin(x)$, $0 < x < \pi$.

(xiii) Find the sum of n terms of the series $1.2^2 + 2.3^2 + 3.4^2 + \ldots$

SECTION "C"

Maximum Marks: 30

Note: Attempt any Three questions of the following. Each question carries equal Marks.

QIII.

(i) if 'x' is such that x^2 and higher powers may be neglected, then show that $\sqrt{((1-3x)/(1+4x))} = 1 - 7x/2$

(ii) Insert 'two' geometric means between 64 and 125.

QIV. Maximize $f(x,y) = 3x + 5y$ subject to the constraints $2x + 3y < 12$, $3x + 2y < 12$, $x + y > 2$, $x > 0$, $y > 0$.

QV.

(i) Solve the following system of equations $x - y + 4z = 4$, $2x + 2y - z = 2$, $3x - 2y + 3z = -3$.

(ii) Show that the points A(4,-2,1), B(5,1,6), C(2,2,-5), D(3,5,0) are coplanar.

QVI.

(i) if $\cos(\theta) = -3/7$ and the terminal ray of θ is in the 3rd quadrant, then find $\sin(\theta/2)$.

(ii) Show that $r_1 r_2 r_3 = rs^2$.